

The energetics of molecular assemblies

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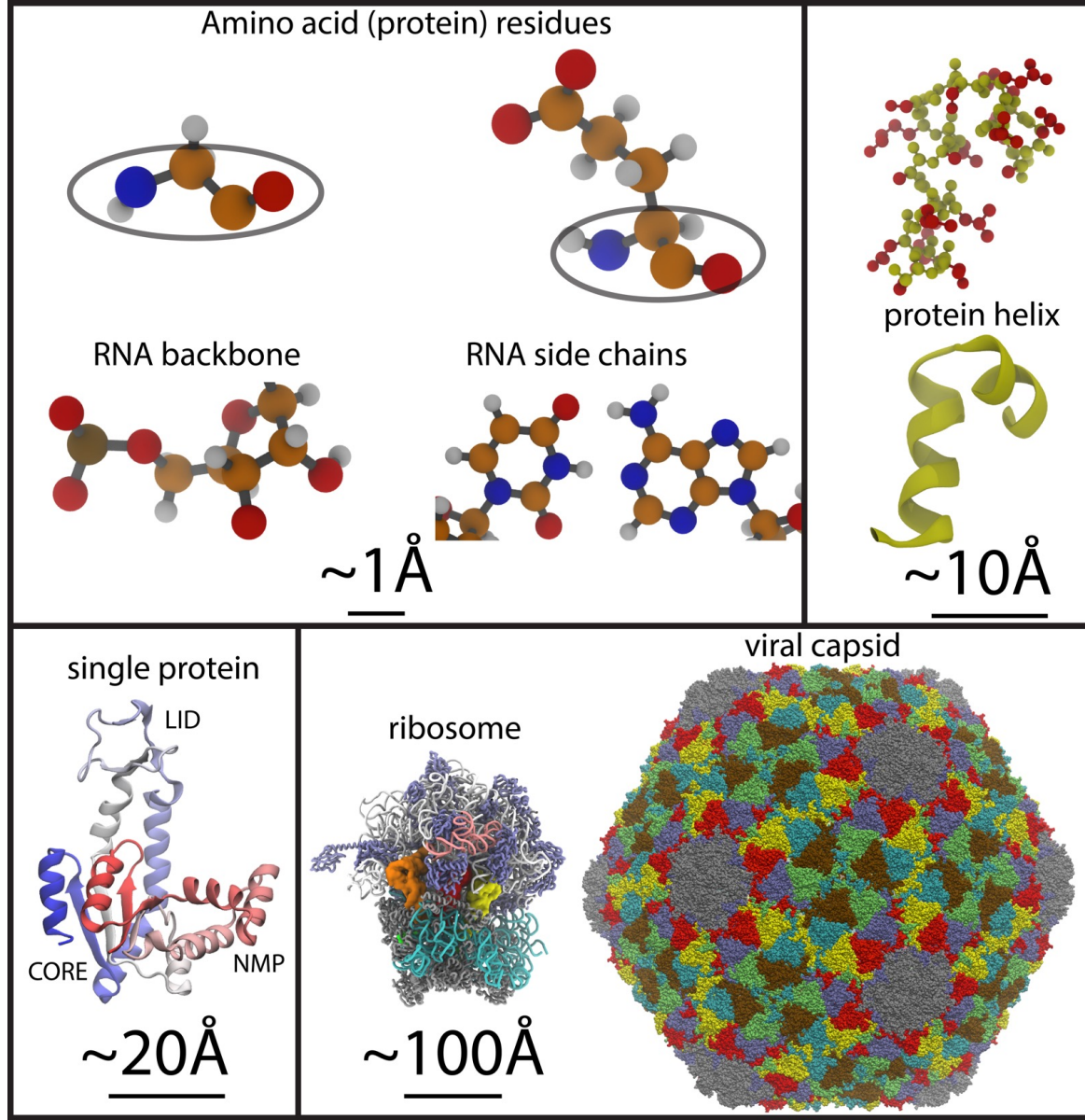
Department of Physics

Northeastern University

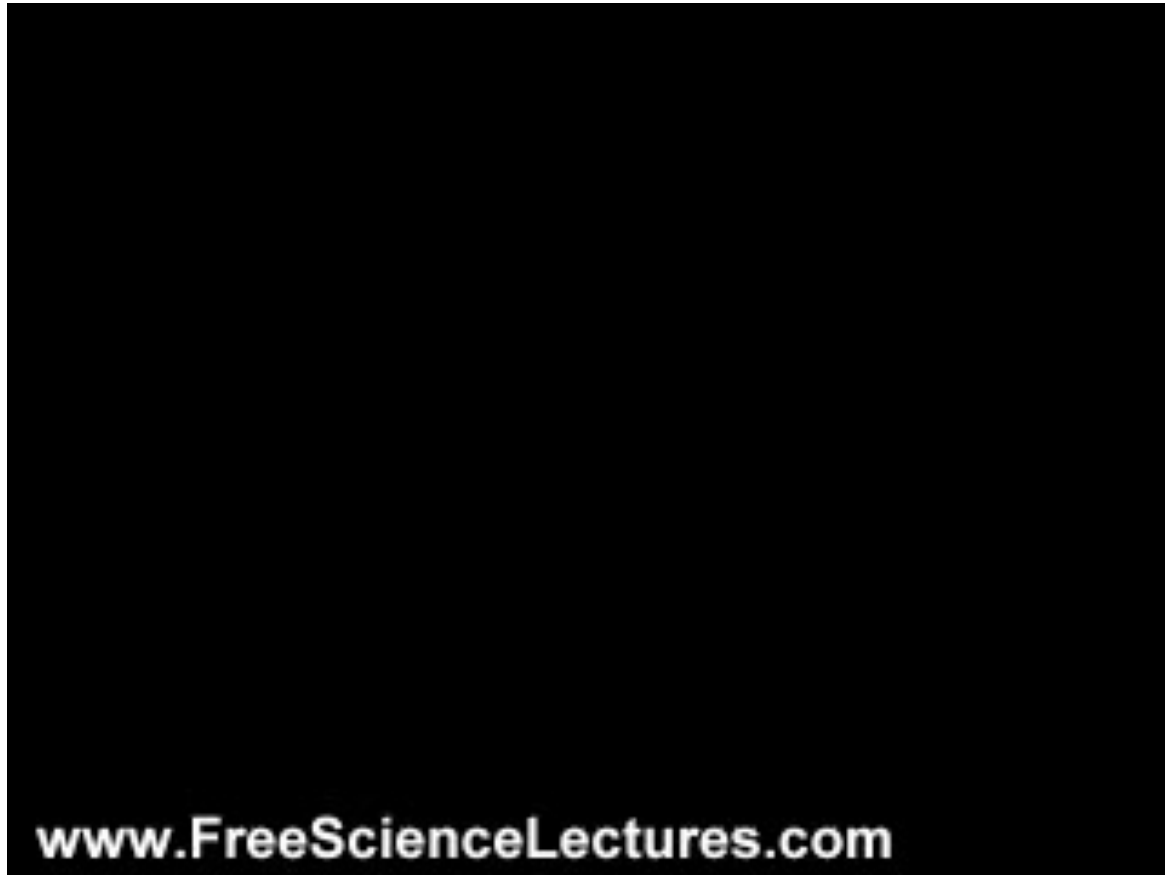
Overview

- Some technical background
 - What is a simulation?
 - How can a simulation be used to study physical properties of molecules?
- Modeling approaches
- Example applications
 - Ribosomes
 - SARS-CoV-2 spike protein

From residues to assemblies



An artist's view of molecular biology



Beautiful, but...

Depicted dynamics are not consistent with physics

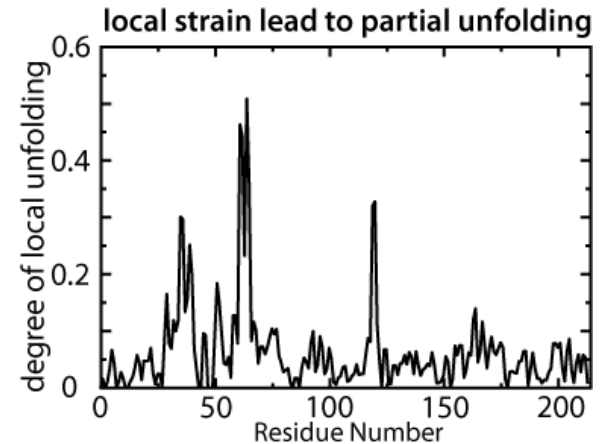
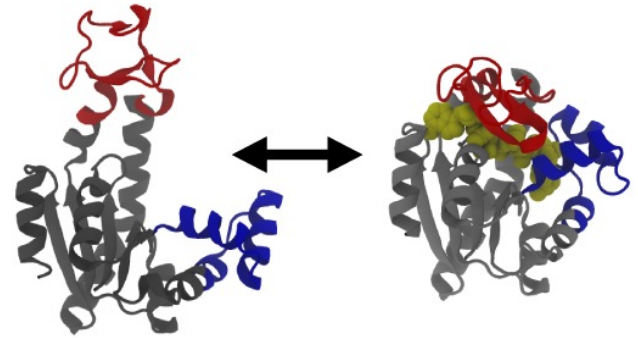
Why? Biopolymers are dynamic!

In our daily lives, using big machines...

When a part breaks, everything stops



Nanomachines (Biomolecules) can break and rearrange dynamically



Lots of energy kicking molecules

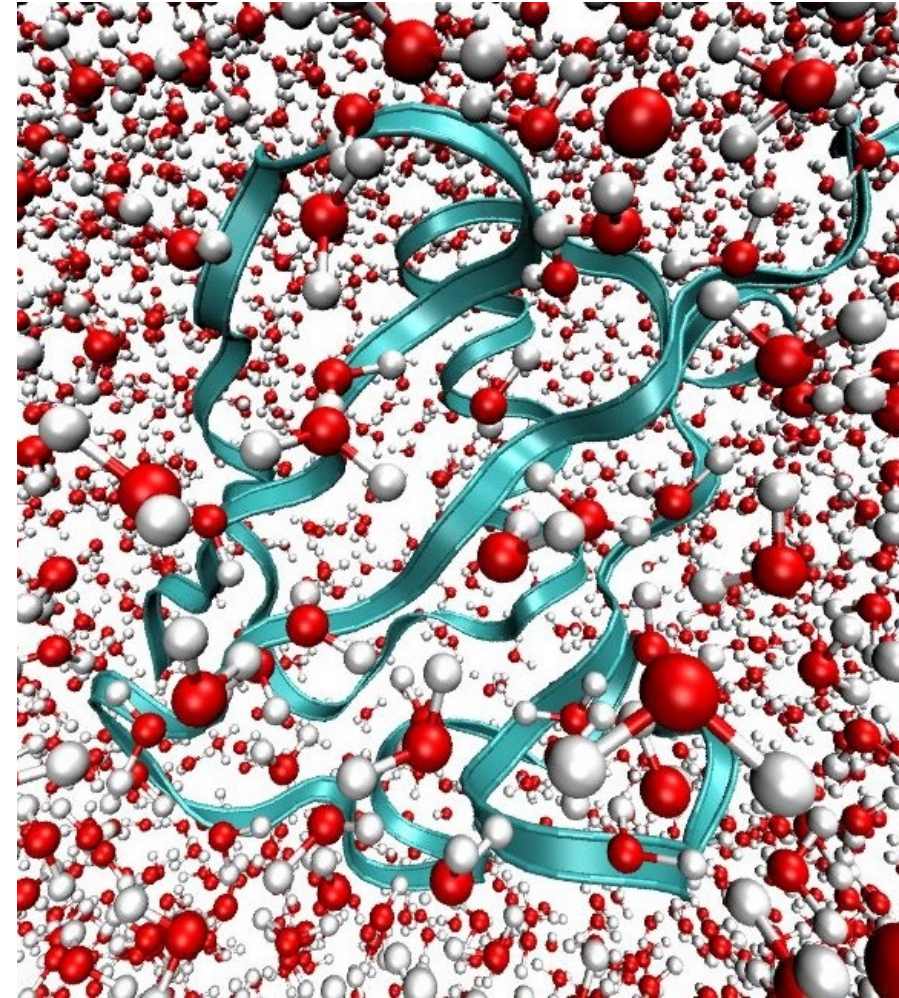
“Tons” of energy constantly being thrown at biomolecules

Enormous fluctuations in energy, and therefore structure

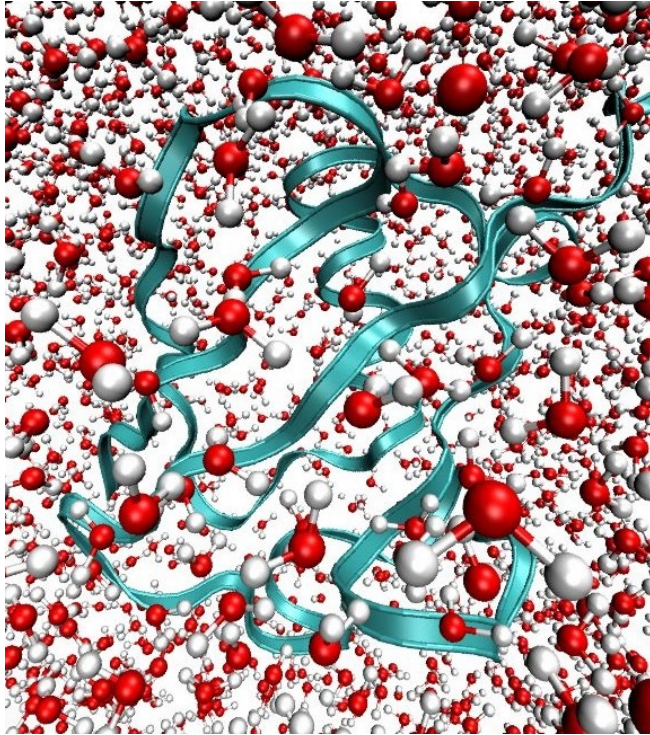
Physical chemistry concepts provide ways to describe the dynamics of systems with large fluctuations

Recall: Individual interactions that stabilize protein (e.g. hydrogen bonds) provide stabilizing energies on the scale of $1 k_B T$

But! Each water molecule has $\sim 1 k_B T$ and the water can interact with the protein



What is a simulation?



Basic physics

$$F = -\frac{d}{dx}U(x)$$

$$F = M \cdot a$$

$$v(t) = v_0 + a \cdot t$$

$$x(t) = x_0 + v \cdot t$$



- Simulations increment the positions and velocities, using small time steps
 - Many many times....
- One calculates the force between all N^2 atom pairs in the simulated system many times

How many calculations go into a simulation?

- Say there are 1000 atoms in a simulation
- Timesteps of 1 femtosecond (10^{-15} seconds)
- How many calculations are required to simulate 1 second of real time?

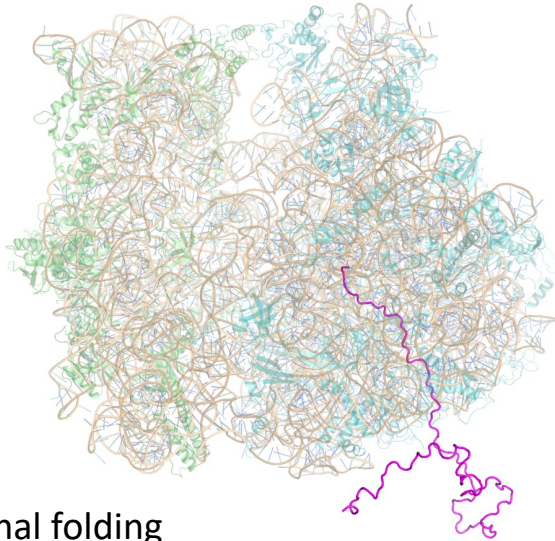
A little math

- 1000 atoms: $1000 * 1000 = 1,000,000$ calculations per timestep
- 1 femtosecond step: 10^{15} steps to simulate 1 second
- Total calculations = $(10^{15} \text{ steps/second}) * (10^6 \text{ calculations/step})$
 - 10^{21} calculations for 1 second of simulated time
 - 1,000,000,000,000,000,000,000 calculations!!!!!!!

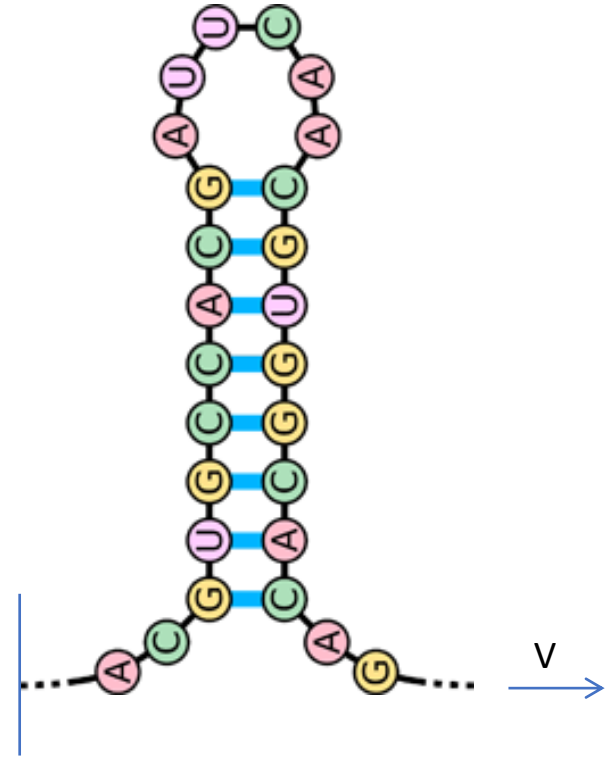
Numerous ways to utilize
simulation techniques

Why perform simulations?

- Kinetic events
- Thermodynamics



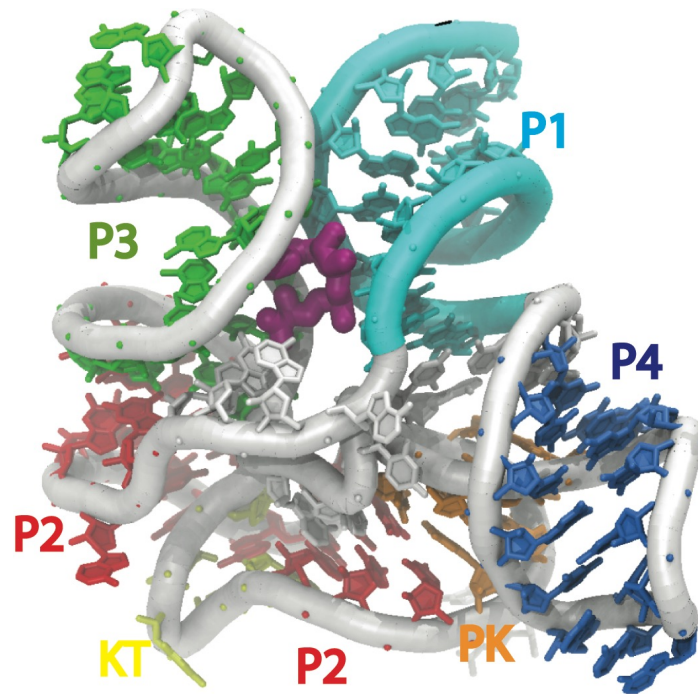
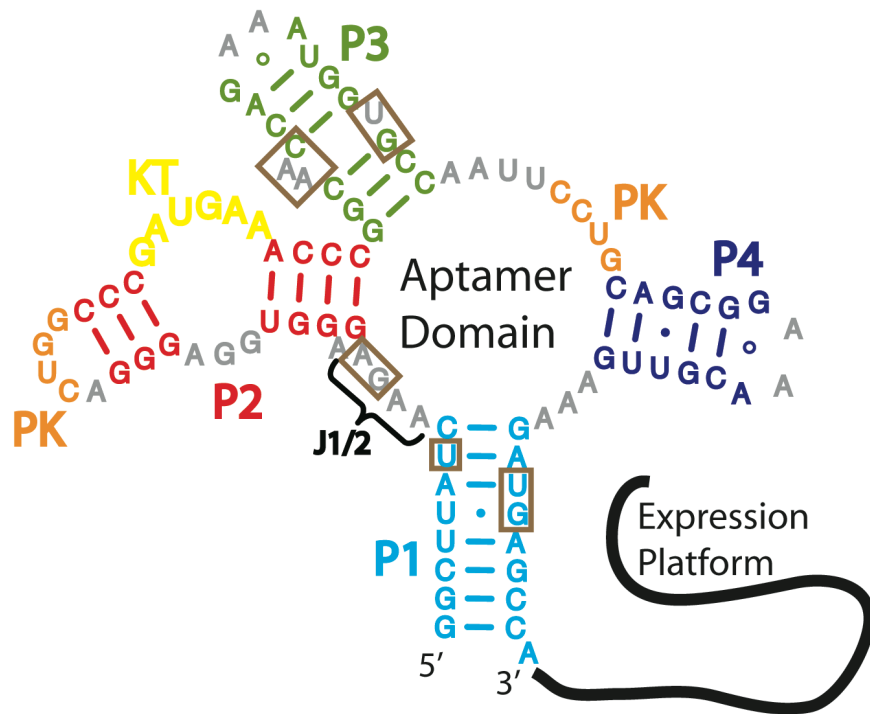
Co-translational folding



Mechanical unfolding

Example of kinetic simulations

Ligand impacts on RNA folding



SAM-1 riboswitch binds SAM while folding.

3' can form alternate helix with terminator (near expression platform)

Competition between the two helices can determine the functional state.

Simulations can suggest a structural perspective of the kinetic properties

Non-local helix formation is key to understanding SAM-I riboswitch function

**P.C. Whitford¹, A. Schug¹, J. Saunders¹, S.P. Hennelly²,
J.N. Onuchic¹, and K.Y. Sanbonmatsu²**

**¹Center for Theoretical Biological Physics,
University of California, San Diego**

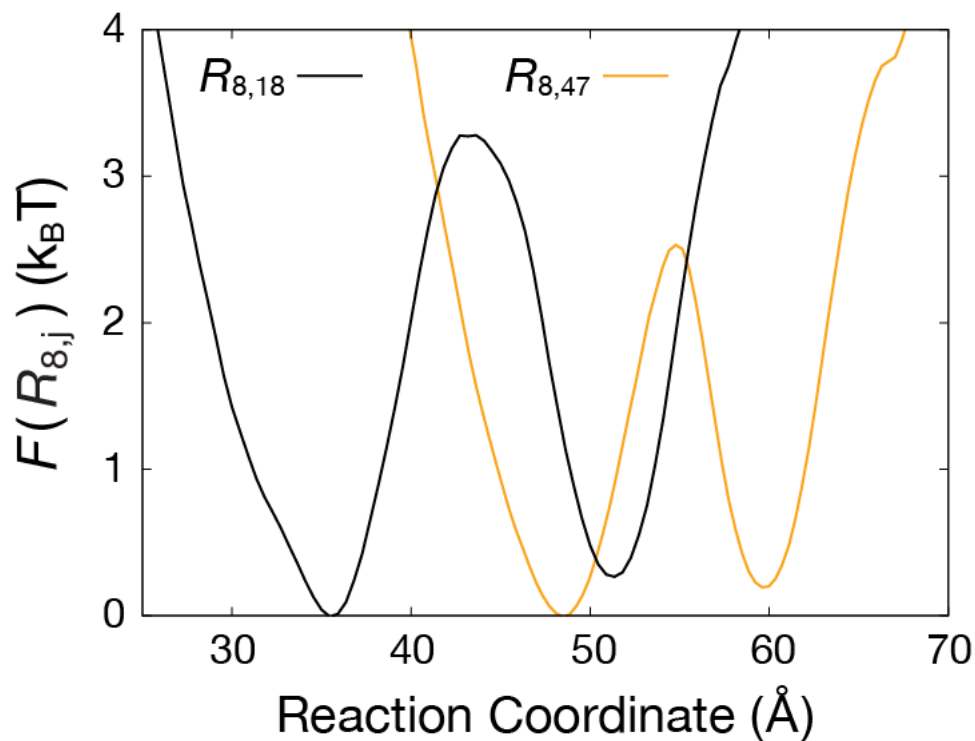
**²T-10 Theoretical Biology and Biophysics
Los Alamos National Lab**

**All-atom Go-model study of ligand binding and
folding of SAM-I riboswitch aptamer domain**

**LANL Coyote Supercomputer
Virginia Tech. System X Supercomputer
For review only -- not for distribution**

Why perform simulations?

- Kinetic events
- Thermodynamics



Thermodynamics

- Thermodynamics may be calculated from the configuration integral Z
 - Very difficult (or impossible) to evaluate directly

$$Z(\beta) = \underbrace{\int d\vec{x}}_{\text{Sum over all configurations}} \underbrace{\exp[-\beta H(\vec{x})]}_{\text{Boltzmann factor}} \quad \beta = 1/k_B T$$

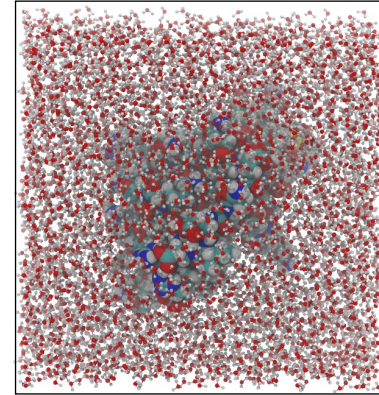
$\vec{X} = (x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, \dots)$

Sum over all configurations Boltzmann factor

$$F(\beta) = -k_B T \log Z(\beta) \quad \text{Free energy}$$

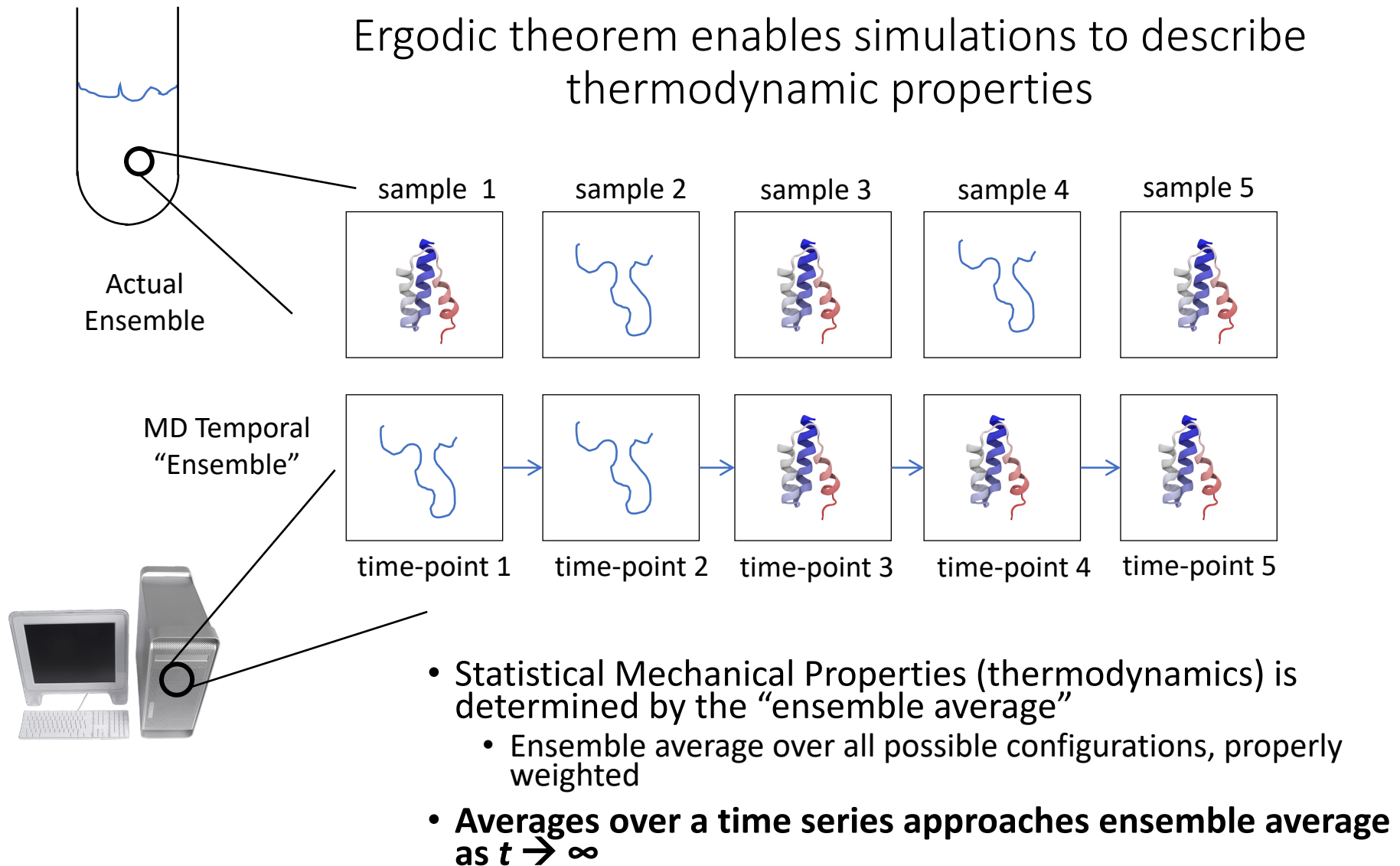
$$F(\beta) = E(\beta) - TS(\beta)$$

Entropy



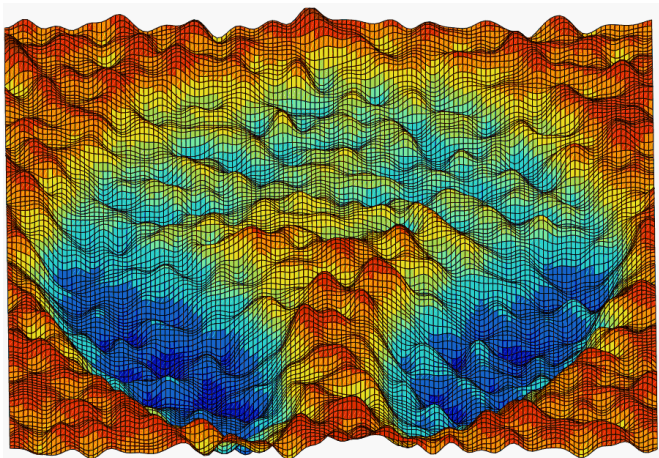
$$P(E) = \Omega(E) \exp(-E/k_B T) \quad \Omega(E) \text{ is the density of states}$$

Ergodic theorem enables simulations to describe thermodynamic properties



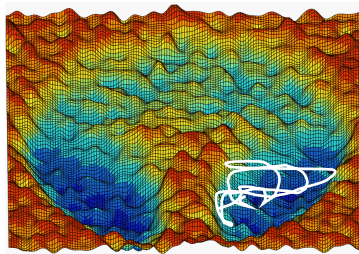
Requirement: The time average must *converge*

Example potential energy surface

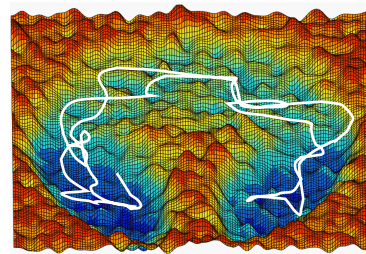


Ren et al. (2005) J. Phys. Chem. B

MD at low T



MD at high T



- Simulations gets stuck in energy minima
- Many advanced sampling algorithms
 - replica exchange
 - umbrella sampling
 - metadynamics, accelerated MD

A simulation may be used
with any model

Different models for different questions

- Structure-based (SMOG) models
 - The spirit - Construct energetically crude models that are sufficiently simple, and sufficiently complete, that specific questions may be asked
 - New models/parameters are built for each system/application
- “conventional” MD (e.g. AMBER, CHARMM)
 - “The goal is to develop the technology to a point where you just have to hit ‘go’ and you will have a complete description of the dynamics” Angel Garcia (paraphrased)
 - “Computational Microscope” Schulten, Amaro and others
 - Completely transferrable parameters
- Coarse-grained “microscope”-style models
 - Transferrable models that reduce degrees of freedom and introduce effective potentials that average over “the details” e.g. MARTINI, MSCG
 - Microscope-like spirit, but intentional omissions of features deemed irrelevant

Brief history of structure-based models

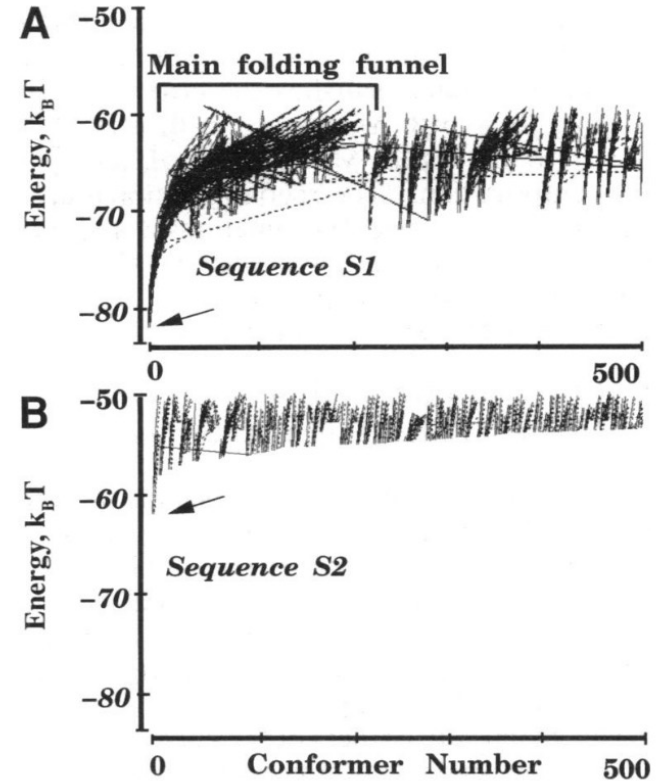
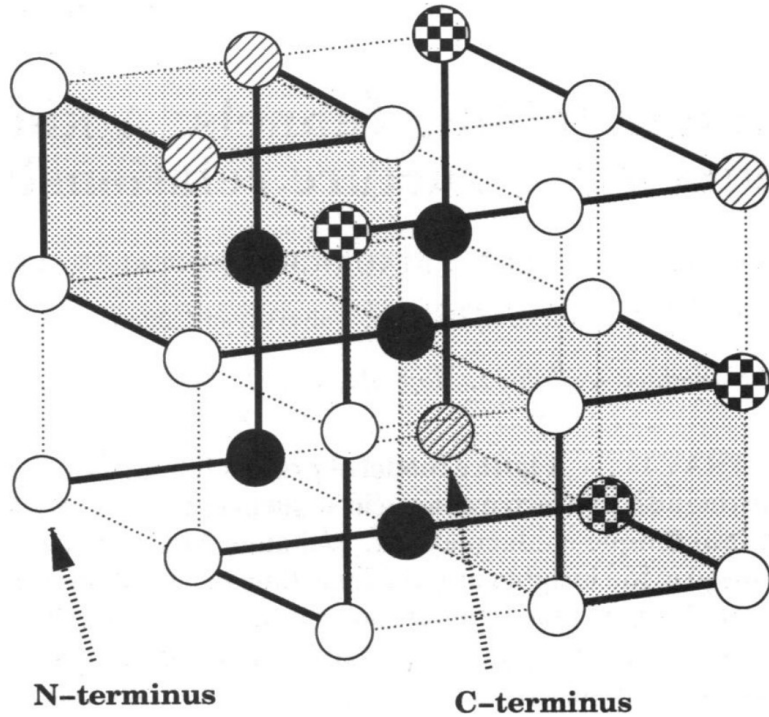
(aka Gō-like models, SMOG models)

- 70s-90s : Lattice-based models for studying protein folding.
Development of energy landscape concepts for studying folding.
- Late 90s, early 2000s : Coarse-grained models with simplified energetics (protein folding focus)
- Early/Mid 2000s : Coarse-grained models with more detailed energetics, extending to conformational rearrangements
- Mid-2000s : All-atom variants with simplified energetics, applications to other complex motions
- 2010+ : Extended to more detailed variants of the models, larger complexes, larger-scale motions

80s : Developing energy landscape principles

- Analytical statistical-mechanical treatments of protein-like models
 - Describing glassy systems, where features could be applied to describe folding energetics
 - Many contributors: Wolynes, Bryngelson, Onuchic, Luthey-Schulten, Thirumalai, Dill, Chan
- Major implication: “Principle of minimal frustration” – Bryngelson and Wolynes 1988 and 1989.
 - Notion that natural proteins have evolved to reduce energetic roughness
 - i.e. energy gap between unfolded and folded ensembles is large, relative to roughness on the landscape
 - Motivated the notion of protein folding “funnels”

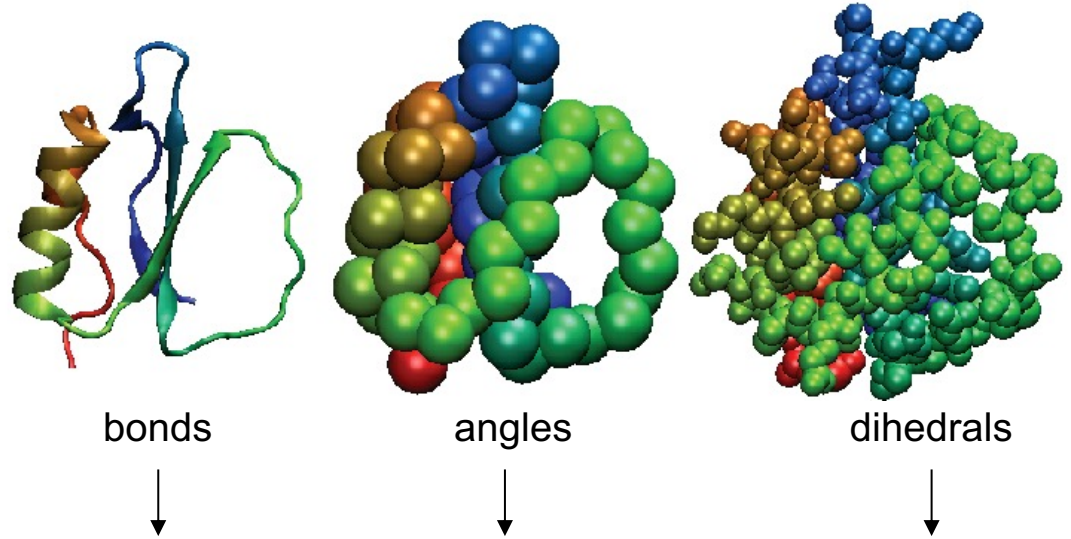
90s: Lattice models and folding funnels



Including a balance between native interactions (i.e. “consistent” interactions) and non-native (inconsistent) interactions to explore the balance necessary for folding to occur on biological timescales

Early 2000s : Off-lattice coarse-grained models

- “Gō” models gained popularity
 - Clementi, Nymeyer and Onuchic JMB 2000
 - Only native interactions are stabilizing
 - “perfect funnel” or “unfrustrate representation of a protein
- Allowed for MD protocols to be applied
- Model built for each protein – typically based on CA atoms, only
- Allowed for specific proteins to be studied and compared against experiments



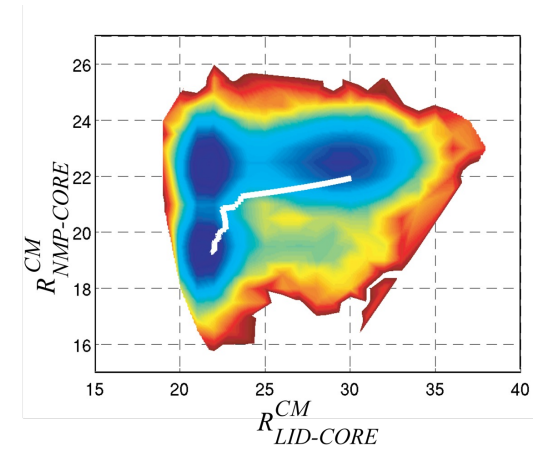
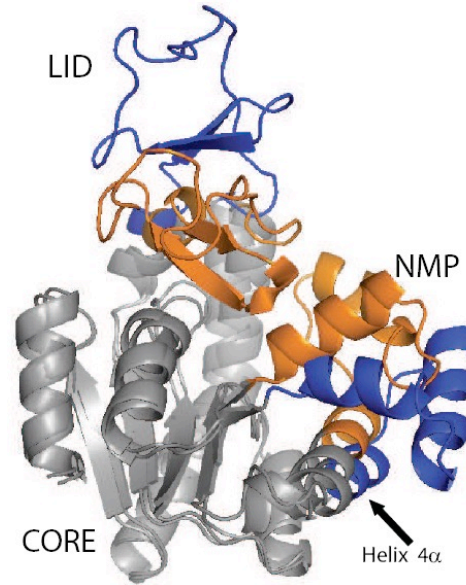
$$H = \sum_{bonds} K_r (r - r_0)^2 + \sum_{angles} K_\theta (\theta - \theta_0)^2 + \sum_{dihedrals} K_\phi^{(n)} [1 - \cos(n \times (\phi - \phi_0))] + \sum_{i < j-3} \left\{ \varepsilon(i, j) \left[5 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 6 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{10} \right] + \varepsilon_2(i, j) \left(\frac{\sigma_{REP}}{r_{ij}} \right)^{12} \right\}$$

Contacts in native state: attractive

Non-contacts: repulsive

Mid-2000s – extending to study conformational changes

- Structure-based models with CA resolution
- Included stabilizing interactions from multiple known conformations
- Allowed one to characterize the interplay between different domains
- Could be used to identify robust aspects of the dynamics

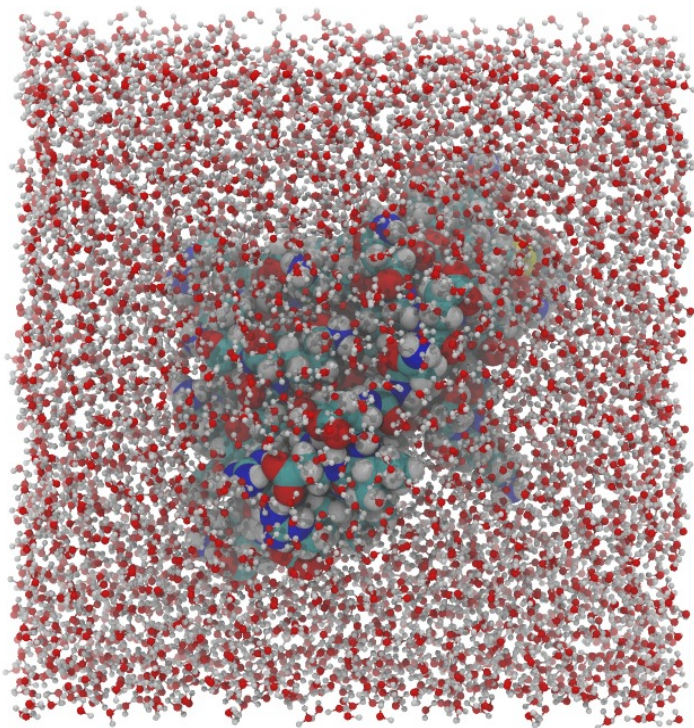


Whitford, Miyashita, Levy & Onuchic, (2007) *JMB* **366**, 1661-1667

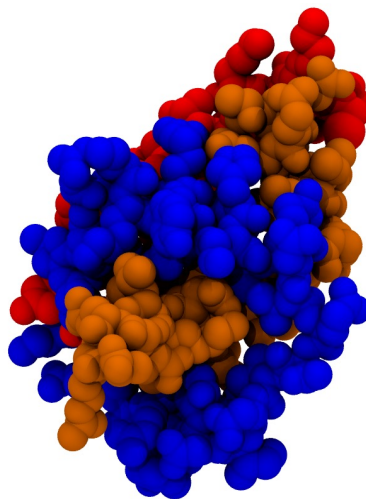
Mid 2000s : All-atom variants

- While CA models were useful toy models, to directly bridge between these types of models and “conventional” MD, all-atom resolution was necessary
- Various all-atom Go-like models were proposed
- We introduced the “SMOG” class of structure-based models
 - Same flavor as the CA models, except all non-H atoms are included
 - Initiated around 2005 through discussions Angel Garcia, who had an earlier variant published along with Clementi and Onuchic
 - Our long-term goal : adopt higher spatial resolution, so that energetics could be incrementally expanded, all the way to CHARMM/AMBER representations
 - Would allow the field to dissect the relative contribution of various factors (H-bonds, electrostatics, ions, pH changes, etc), term-by-term

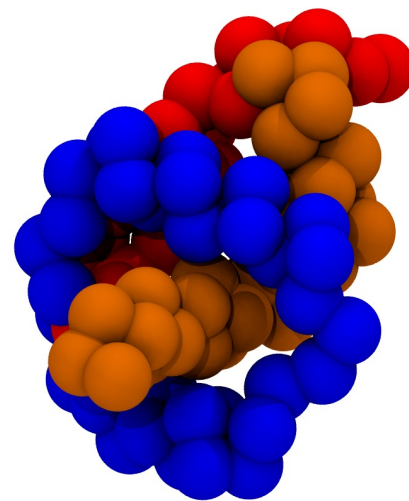
Explicit-solvent model



All heavy atoms



C-alpha atoms

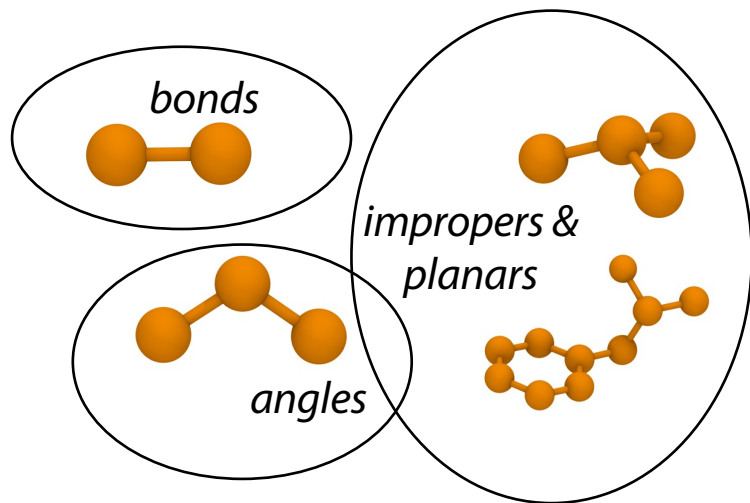


←—————→
High Structural Detail Low

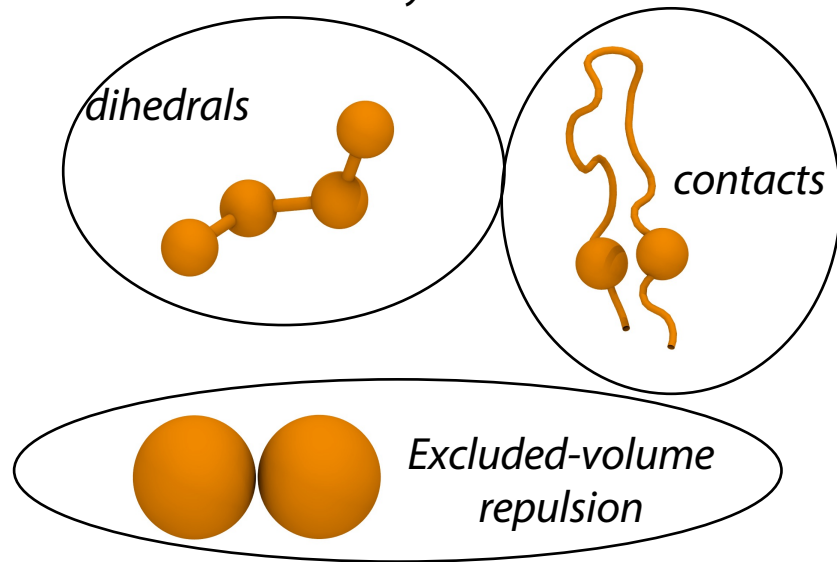
Define the native state to be stable

$$V_{AA}(\vec{\mathbf{x}}, \vec{\mathbf{x}}_0) = \sum_{\text{bonds}} \epsilon_b (r - r_0)^2 + \sum_{\text{angles}} \epsilon_\theta (\theta - \theta_0)^2 + \sum_{\text{impropers/planar}} \epsilon_\chi (\chi - \chi_0)^2 + \sum_{\text{backbone}} \epsilon_{BB} F_D(\phi - \phi_0) + \sum_{\text{side chains}} \epsilon_{SC} F_D(\phi - \phi_0) + \sum_{\text{contact map}} \epsilon_C^{ij} C(r_{ij}, r_0^{ij}) + \sum_{\text{non-contacts}} \epsilon_{NC} \left(\frac{r_{NC}}{r_{ij}} \right)^{12}$$

Local structure propensity



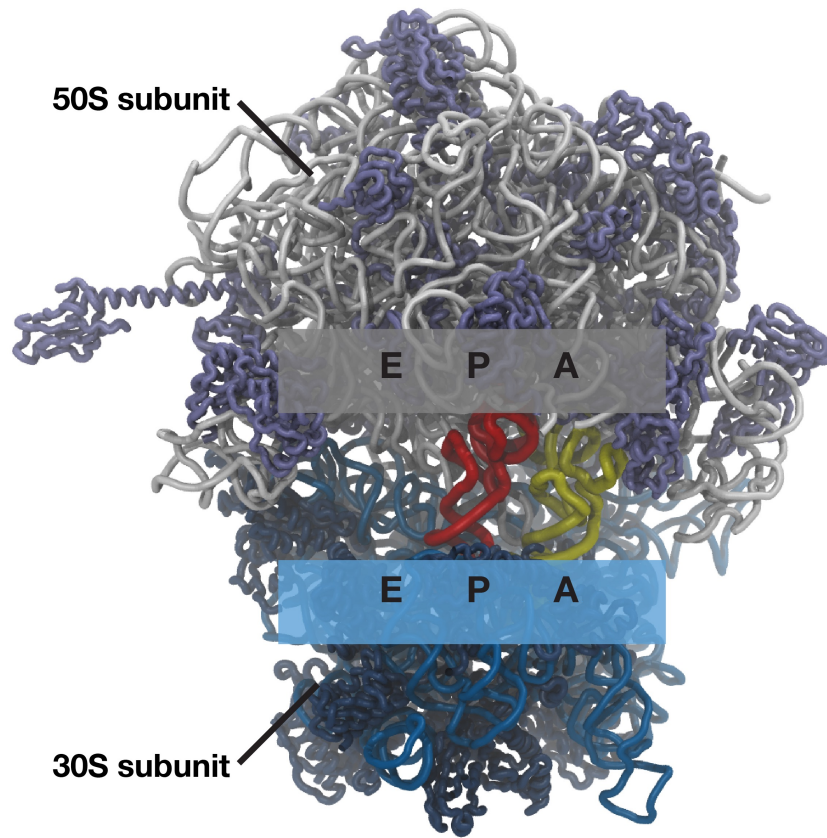
Tertiary structure



Extending folding ideas to study molecular machines

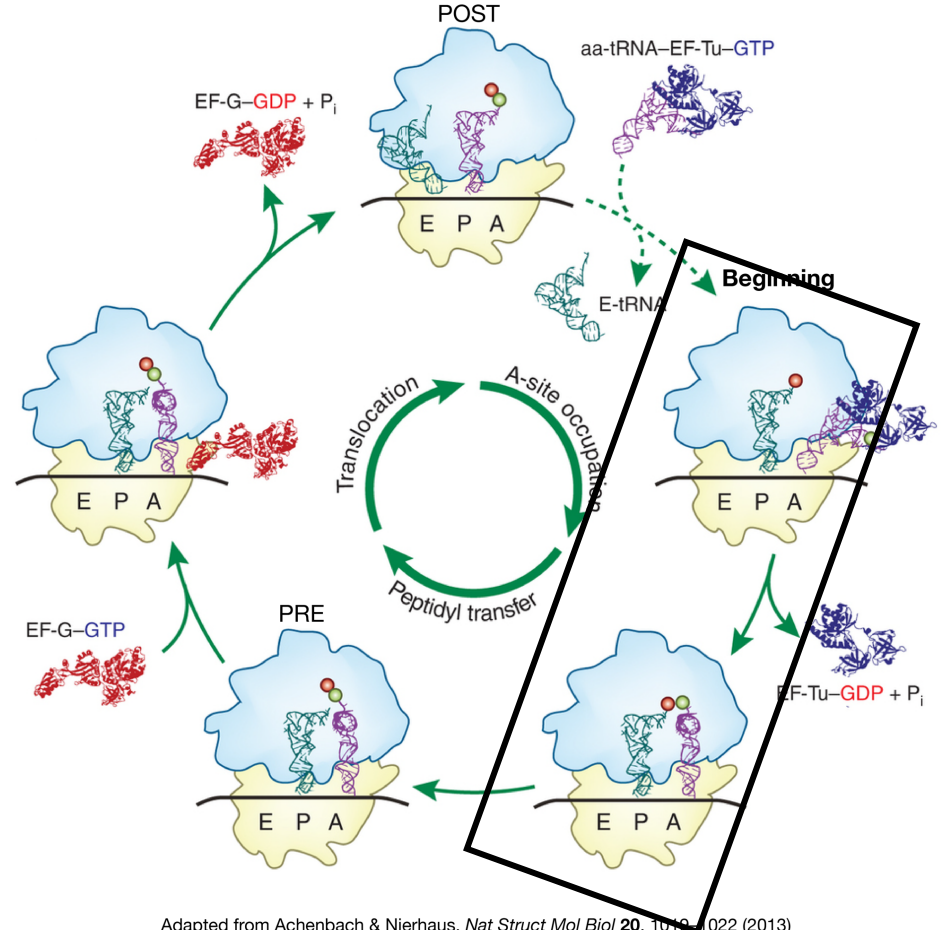
- Examples
 - The ribosome
 - SARS-CoV-2 Spike protein

The Ribosome



Over 50 proteins
3 RNAs w/ ~4000 nt
~150,000 atoms

PDB ID: 4V6F
Jenner et al. (2010)

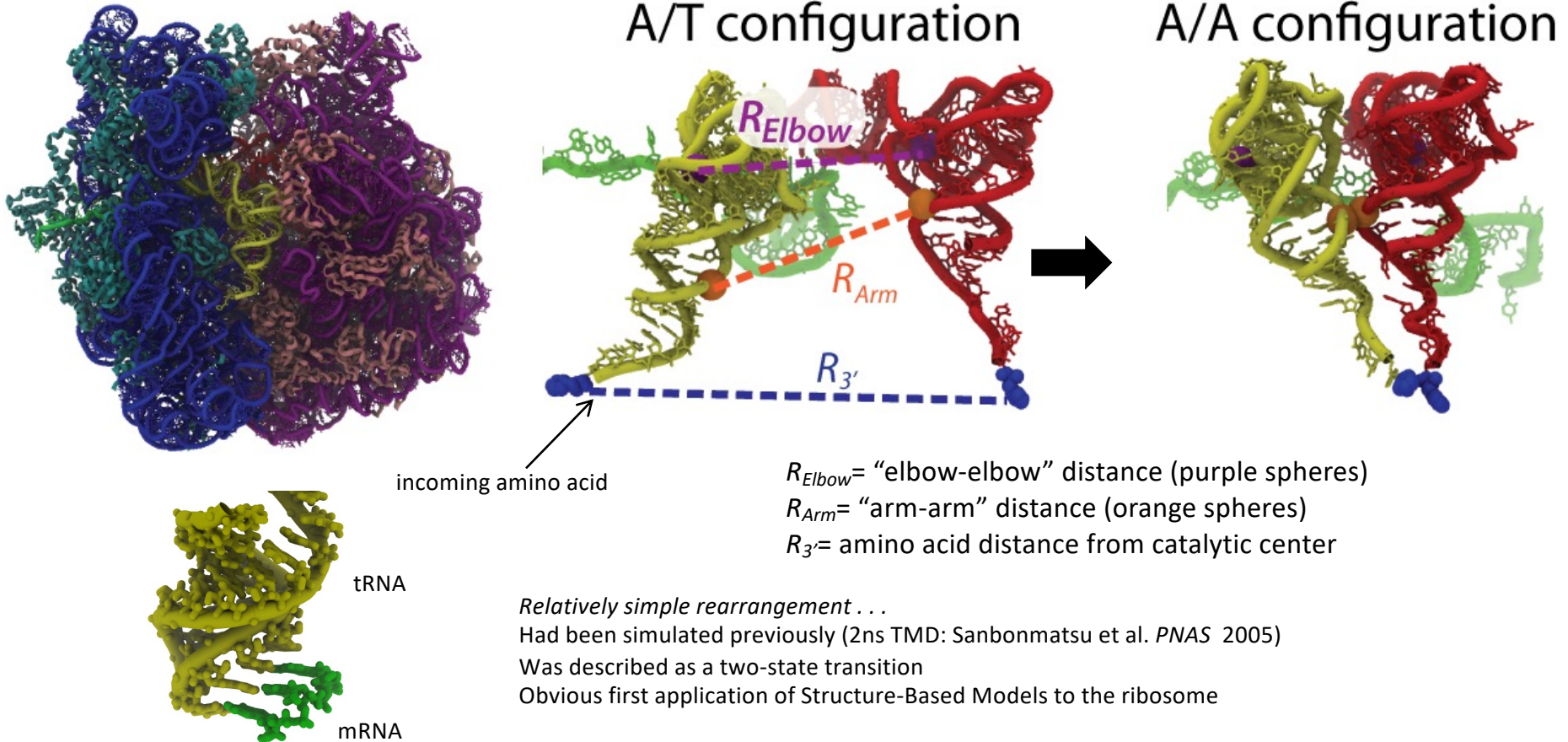


Adapted from Achenbach & Nierhaus, *Nat Struct Mol Biol* **20**, 1019-1022 (2013)

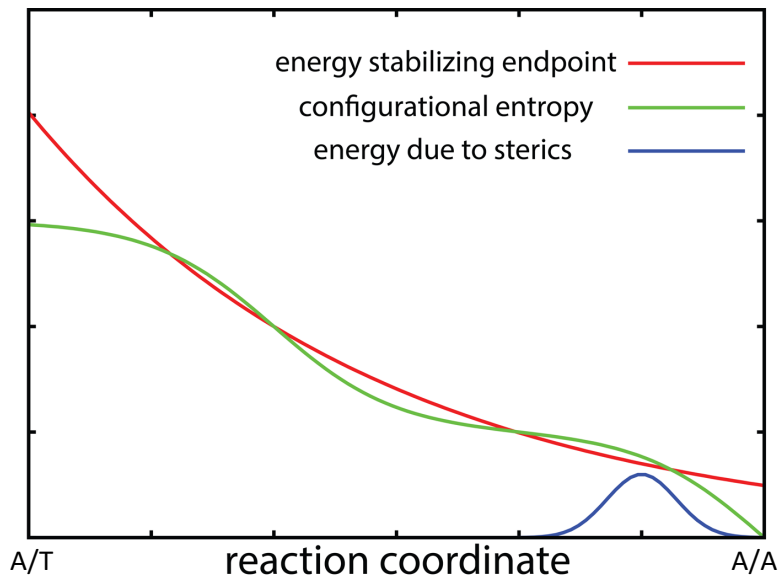
tRNA entry: Accommodation

tRNA accommodation involves a ~ 90 Å displacement of new amino acid into PTC

Accommodation may be important for fidelity



What can we learn from a simple model?

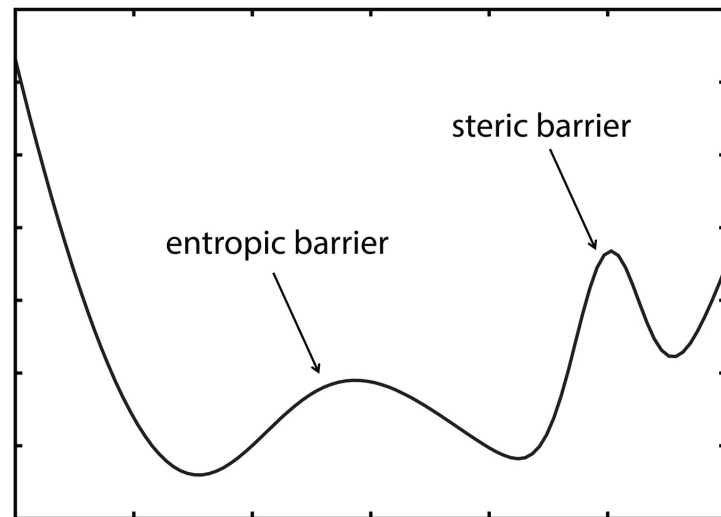


The Structure-based (SMOG) Model:

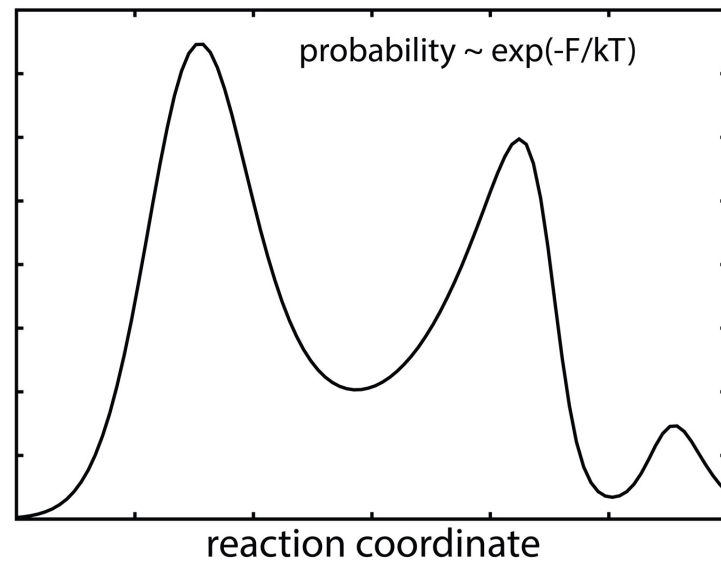
- Include all heavy (non-hydrogen atoms) atoms
- A/A state is the lowest effective-enthalpy state
- Consistent with the notion of tRNA as a 'molecular spring' (Frank et al. *FEBS* 2005), where tRNA has strain energy

All-atom SMOG models (PCW et al. *Proteins* 2009)
Reviewed in Levi, Noel, PCW *Methods* (2019).

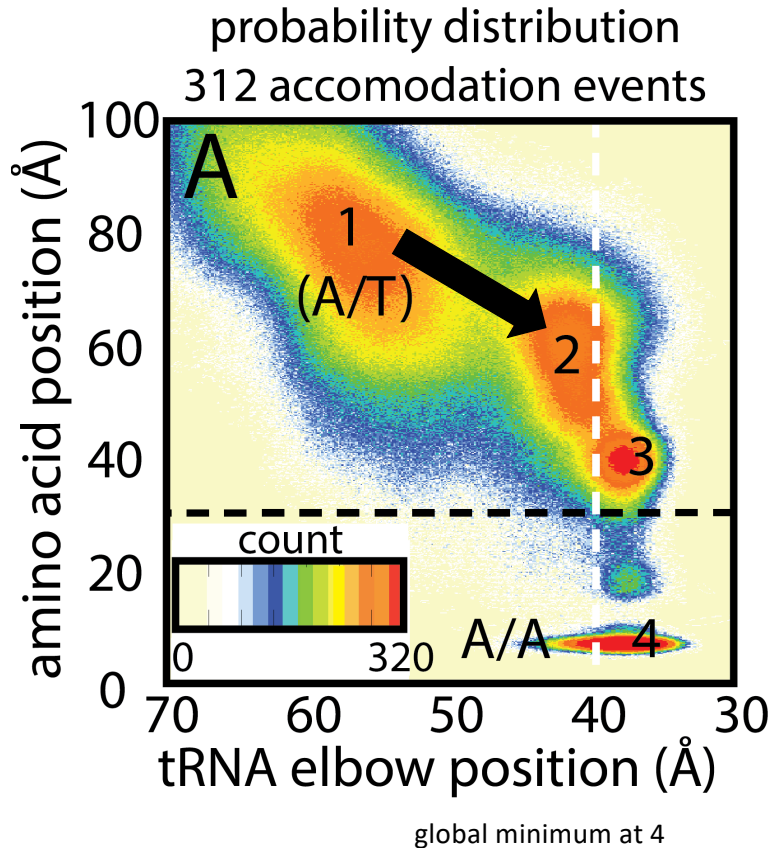
free energy = total energy - $T \cdot \text{entropy}$



probability $\sim \exp(-F/kT)$



Proof of principle: Sterics can limit accommodation



Peaks result from sterically-induced barriers
Predicted a multi-step process, due to sterics

tRNA accommodation with a SMOG model

Representative trajectory described in:
Whitford et al. *RNA* 16, 1196-1402, 2010.

NMCAC Encanto Supercomputer

Can EF-Tu directly impact accommodation?

Observations

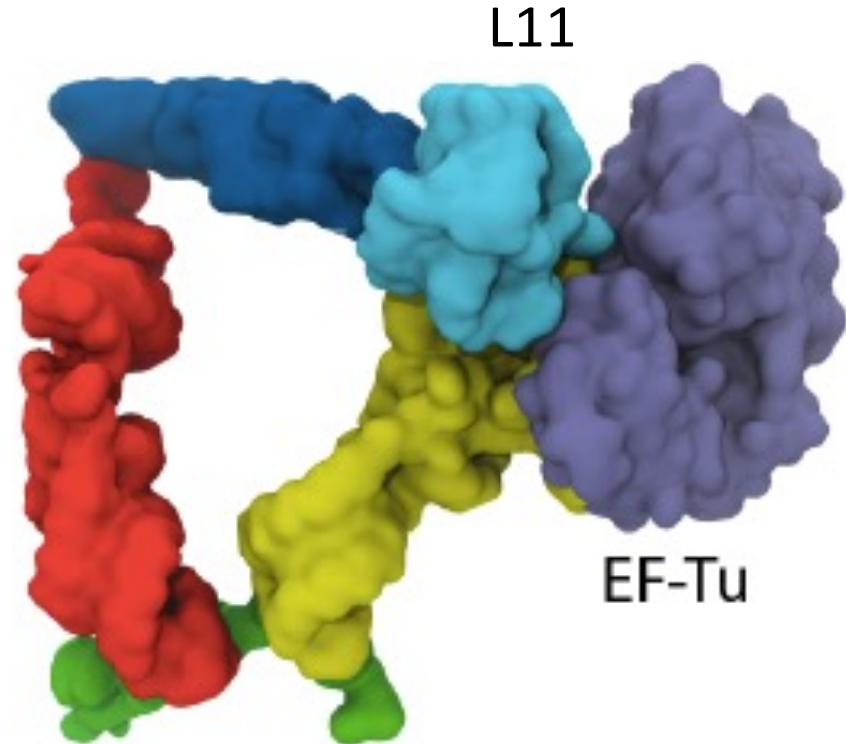
- EF-Tu delivers tRNA to the ribosome
- Experiments suggest EF-Tu dissociation can be slower, or faster than accommodation of aa-tRNA (Pape et al, EMBO J 1999) and that aa-tRNA accommodation rates can significantly depend on conditions (Johansson et al. *Mol Cell* 2008).
- Models predict there are at least two conformational substeps: elbow accommodation and 3'-CCA accommodation

Question

Can the rate of EF-Tu dissociation from the ribosome impact accommodation rates?

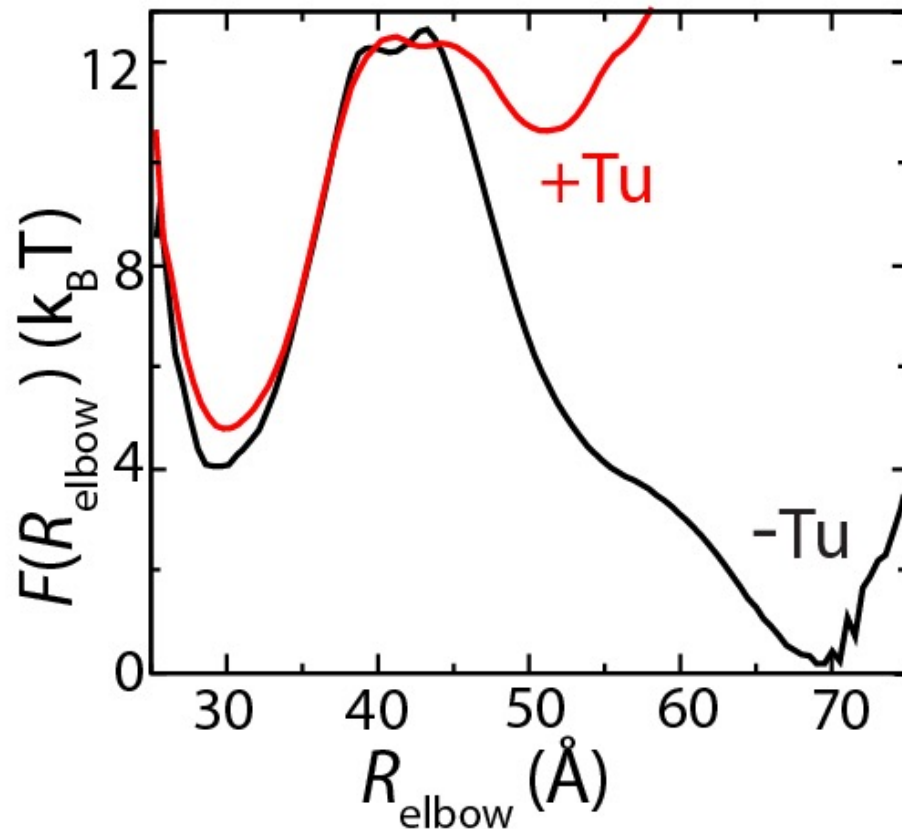
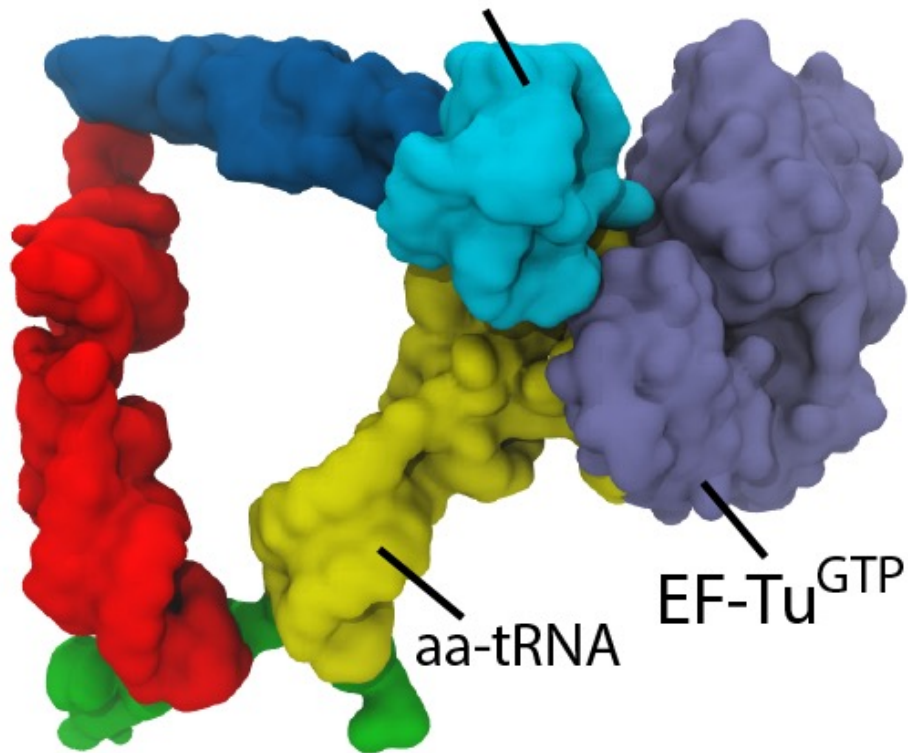
Strategy

Use SMOG models to identify influence that EF-Tu can have on accommodation through steric and entropic contributions



EF-Tu can accelerate accommodation

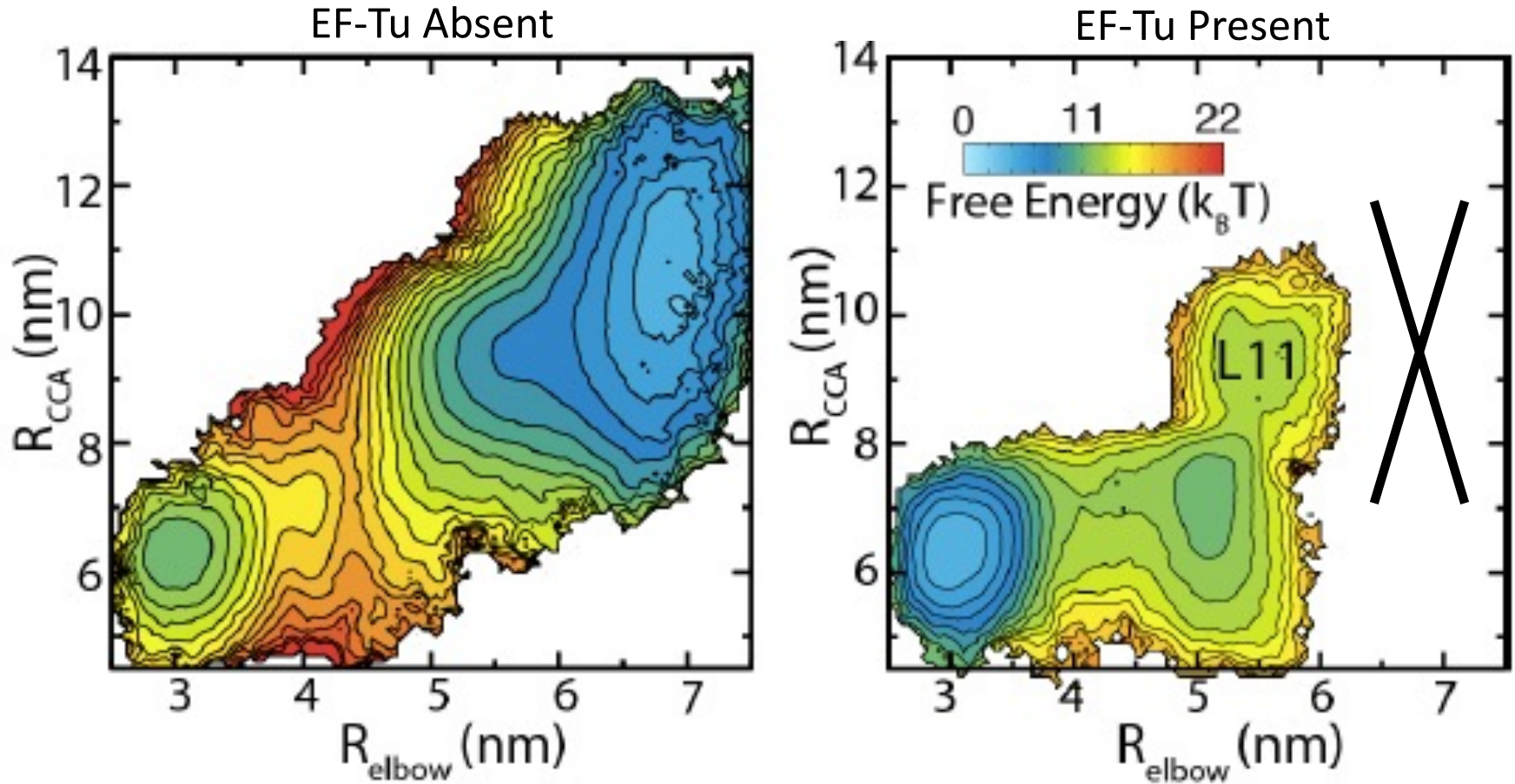
L11 stalk



Red: EF-Tu bound to ribosome, tRNA released
Black: EF-Tu absent

Barrier much larger in the absence of EF-Tu
(orders of magnitude slower kinetics)

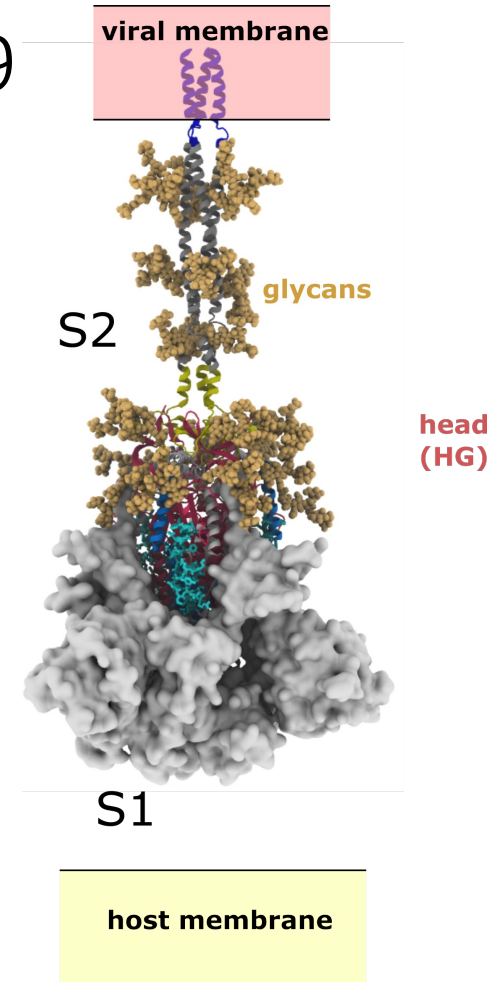
Sterics of EF-Tu reduce accessibility of A/T ensemble

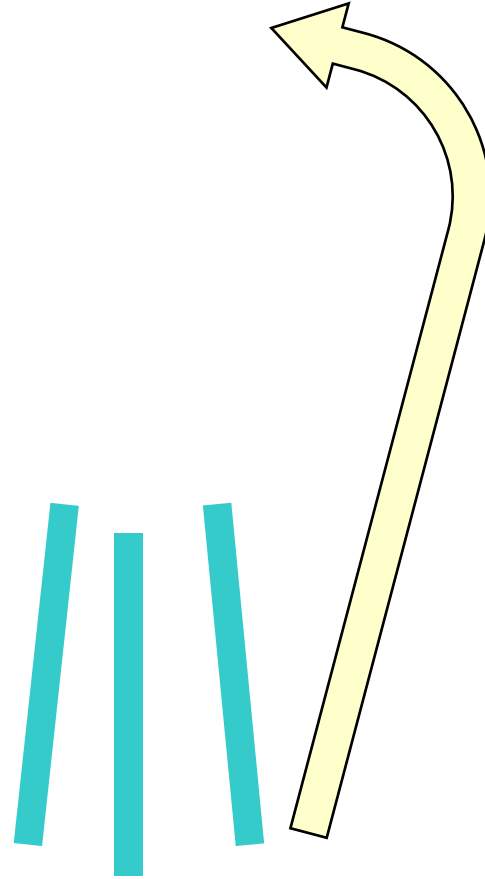
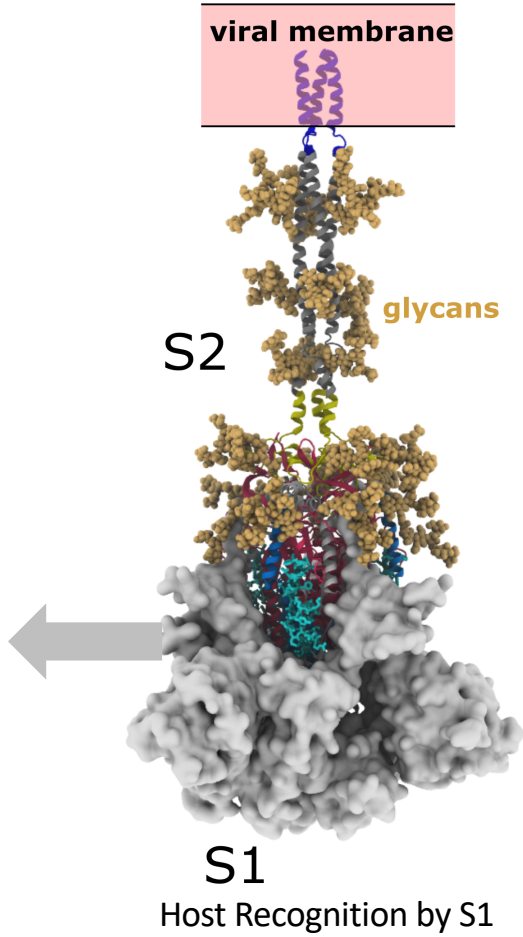


- EF-Tu restricts tRNA mobility, which decreases the configurational entropy of the A/T ensemble
- By limiting accessible phase space, forward barrier is lowered

Understanding COVID-19

- SARS-CoV-2 Spike protein (right) recognizes human cells and allows the virus to enter
- Using simulations to tackle Spike from multiple angles
 - **Effort 1: Biophysical characterization may help identify the “Achilles’ Heel” of the Spike**
 - Effort 2: Simulation can guide development of new vaccines



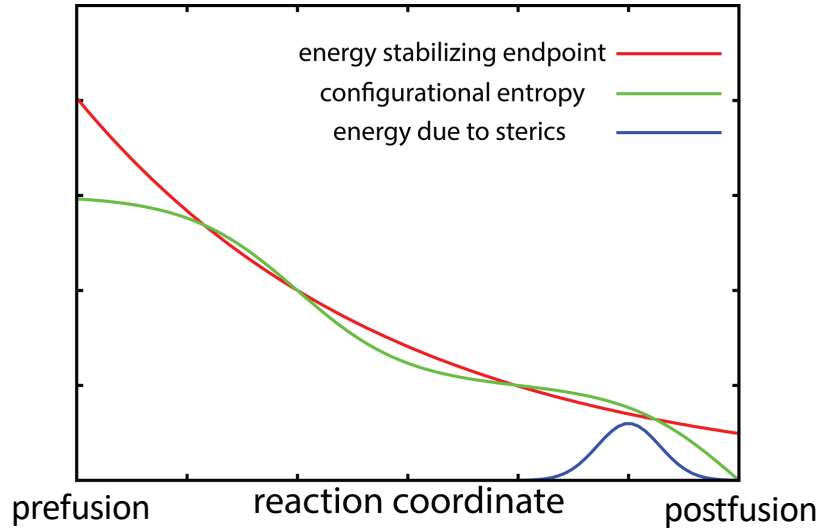


Membrane fusion

Simulating the Spike during membrane fusion

- Experiments have only provided structures before and after membrane fusion
- Membrane fusion involves a ~ 20 nm rearrangement of the Spike
- By simulating the transitions, we may find which features are more exposed during the rearrangement
- Exposed regions may be targeted with novel vaccines

Using a simplified model

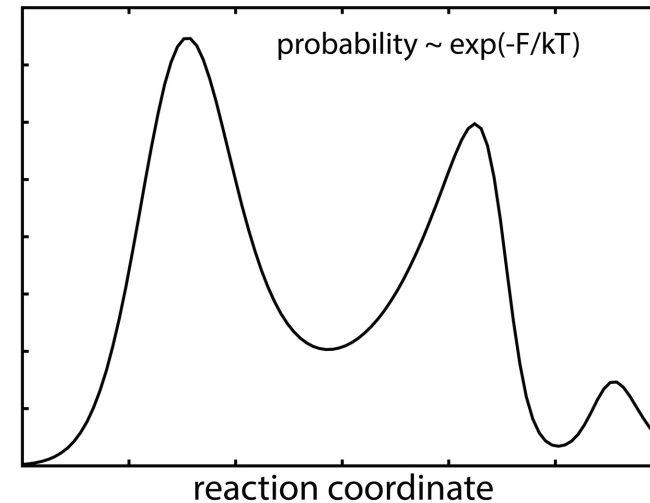
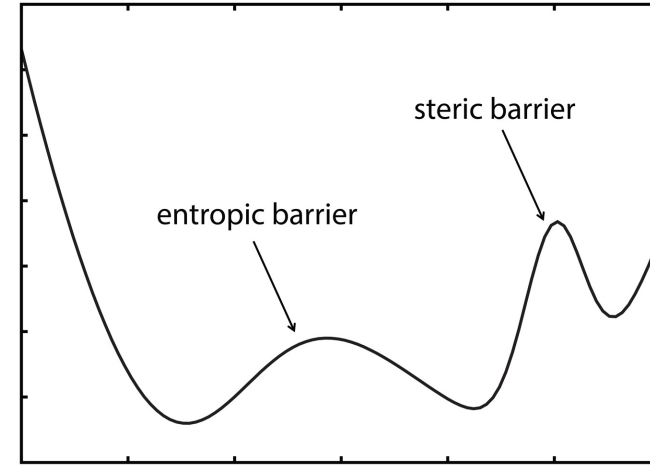


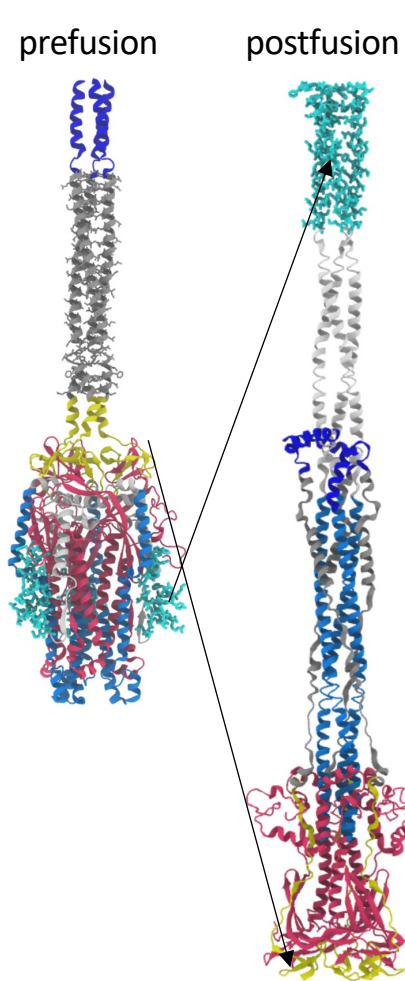
The Structure-based (SMOG) Model:

- **Includes all heavy (non-hydrogen atoms) atoms**
- postfusion state is the lowest effective-enthalpy state
- All-atom SMOG models (PCW et al. *Proteins* 2009)

Reviewed in Levi, Noel, PCW *Methods* (2019).

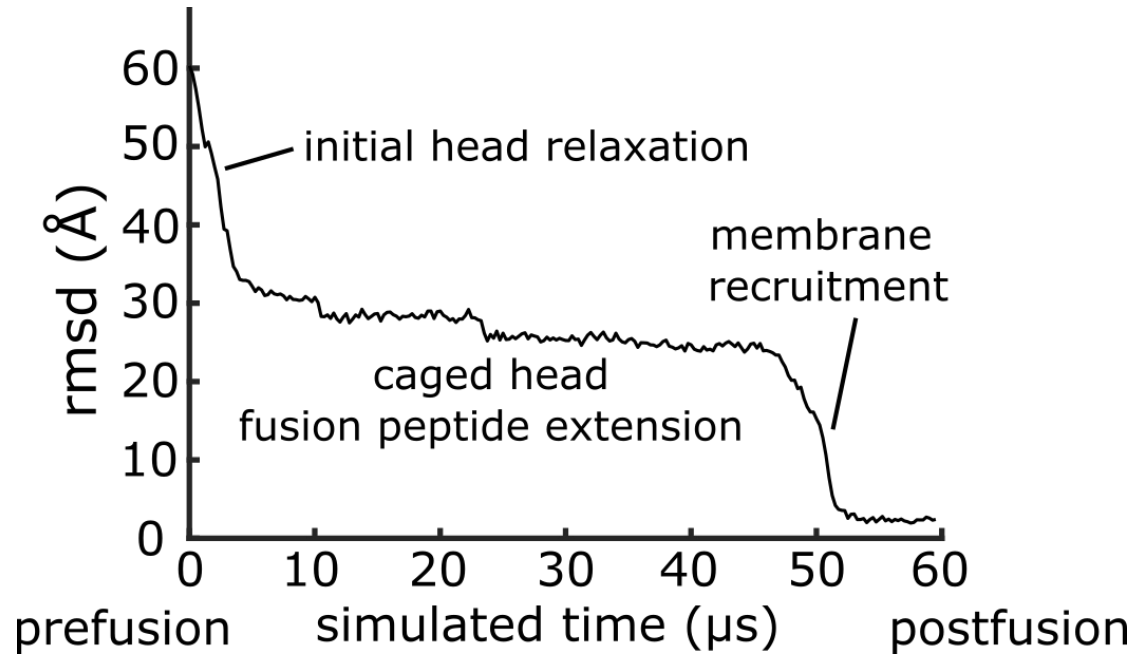
$$\text{free energy} = \text{total energy} - T \cdot \text{entropy}$$





Simulating Rearrangements in Spike

HG	706
FP	816
CR	855
HR1	939
HG	1012
LR	1099
HR2	1150
TM	1213
	1233

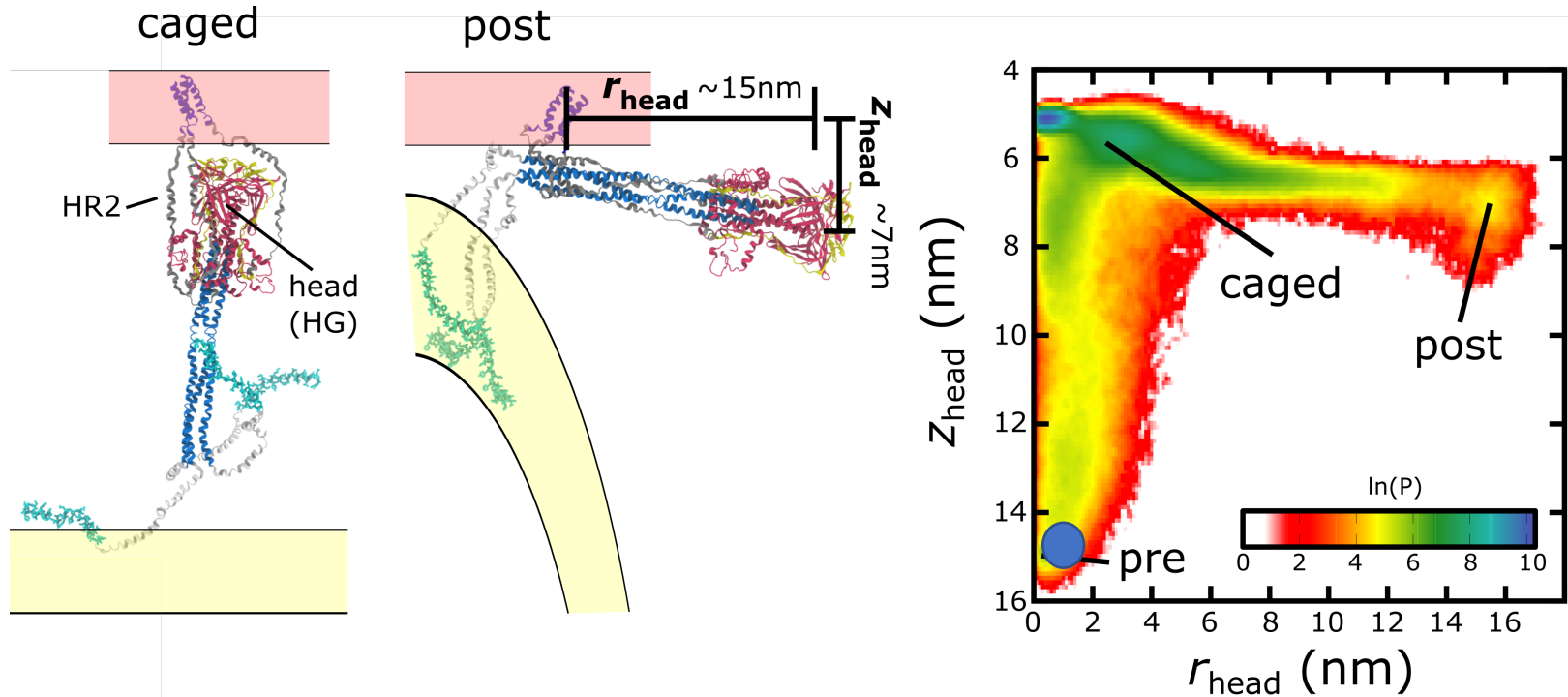


Simulations capture the full rearrangement process
Each atom moves (on average) ~6 nm (up to ~30 nm)

Thanks to AMD HPC Fund, we can now run >600 runs at a time (each with a MI50 GPU)

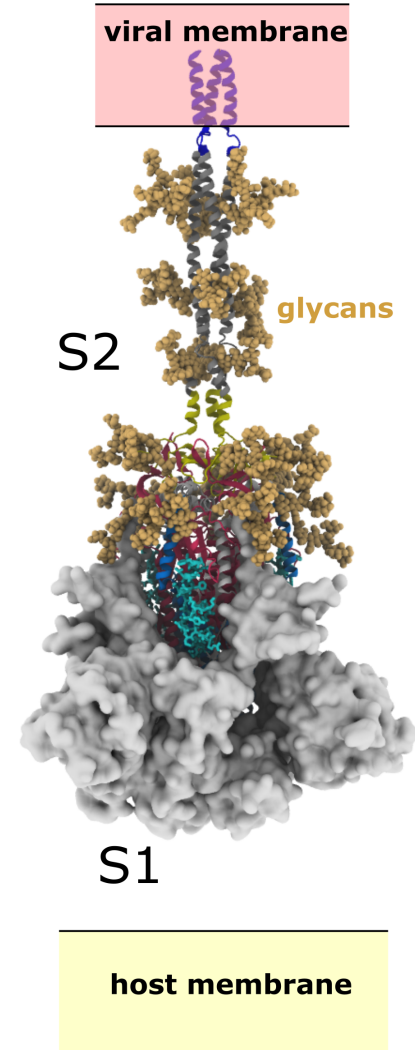
Dodero-Rojas, Onuchic and Whitford *eLife* 2021

Statistical Properties from 1000 events



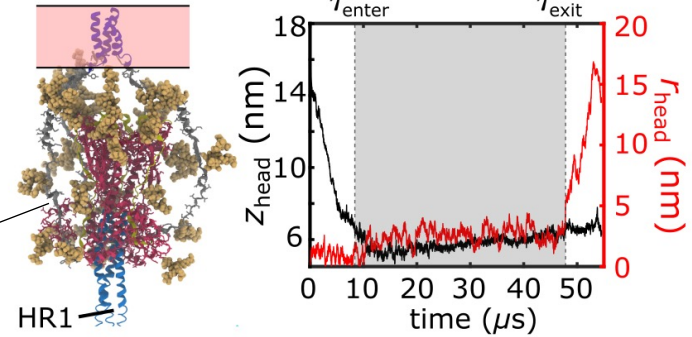
What about the glycans?

- Glycans are simple sugars (gold) that coat the Spike protein
- Others have suggested that glycans can protect the spike from recognition by the immune system
- Our simulations suggest the system can become caged for an extended period of time
 - *Perhaps glycans also influence this process*

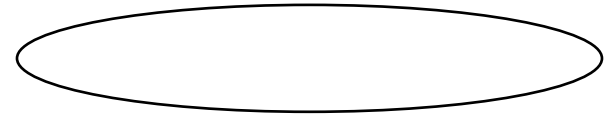


Glycans slow down the rearrangement

The caged state appears to engage the glycans



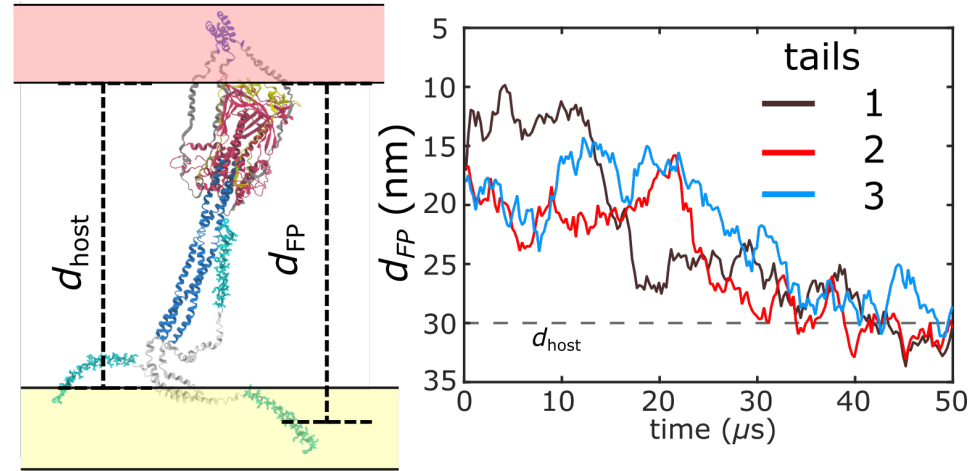
Spike can remain caged for long times when glycans are present



In absence of glycans, caged state is bypassed!

Glycans extend reach of fusion peptides

In order to infect a cell, the fusion peptides (tails) must reach the host (yellow)



The chance of reaching the host (P_{capture}) is much larger when glycans are present



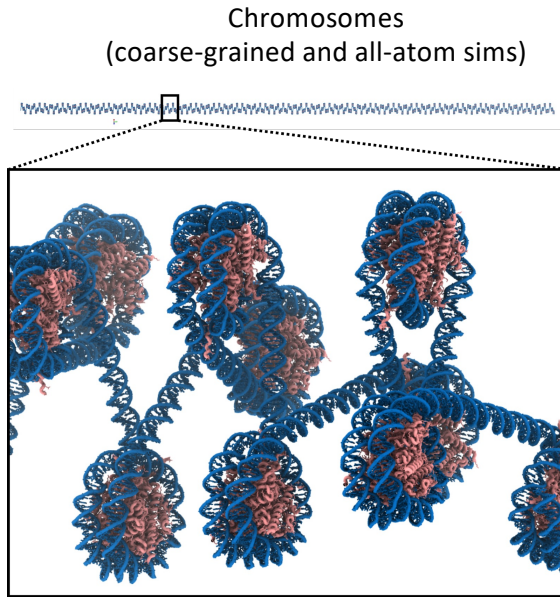
Simulation of SARS-CoV-2 Spike-Protein-Mediated Membrane Fusion

Video 1 for
"Sterically-confined rearrangements of
SARS-CoV-2 Spike Protein Control
Cell Invasion" (2021)

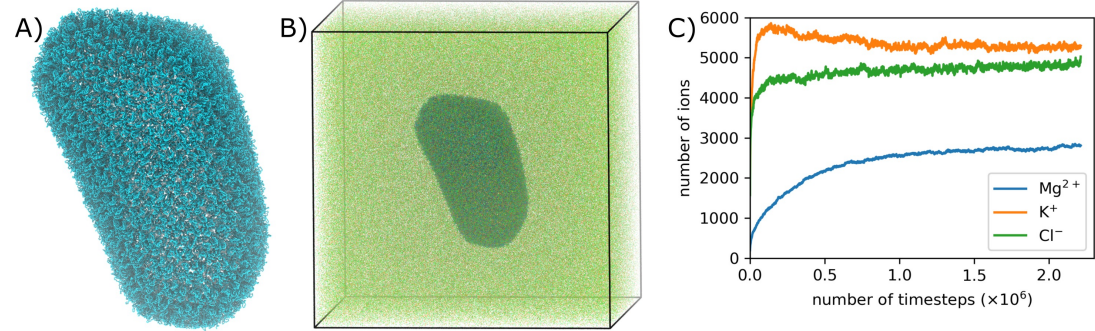
Esteban Doderro-Rojas, José N. Onuchic,
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Rice University
Northeastern University

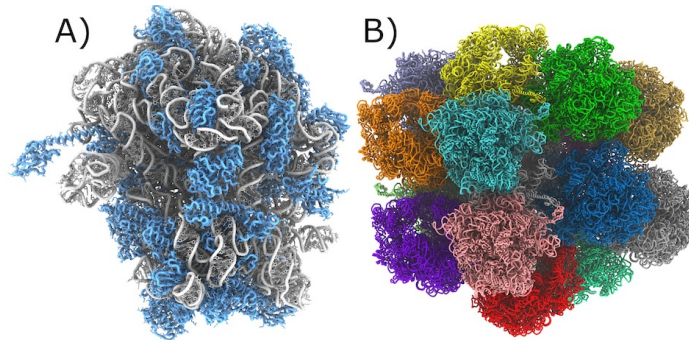
Expanding to other complex assemblies



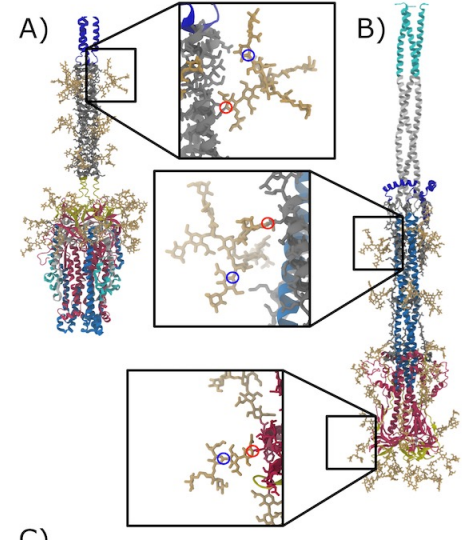
HIV-1 Capsid – explicit ions and electrostatics



Ribosomes



COVID Spike Protein



Our models are supported by
Gromacs for large systems (millions of atoms on thousands of cores)
OpenMM (<100k atoms on single GPUs)
NAMD and LAMMPS